

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Winter 2018****Course:** B. Tech in Mechanical Engineering**Sem:** III**Subject Name:** Thermodynamics**Subject Code:** BTMEC305**Max Marks:** 60**Date:** 10 Dec 2018**Duration:** 3 Hr.**Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following six questions.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculator is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/ CO)	Marks
Q.1 Solve the following:		
A) State Zeroth law of thermodynamics. Differentiate between point functions & path functions and write at least two examples of each.	(CO-1)	6
B) With the help of simple diagram, explain working principle of thermocouple and state its advantages. A platinum resistance thermometer has a resistance of 2.8 ohm at 0°C and 3.8 ohm at 100°C. Calculate the temperature when the resistance indicated is 5.8 ohm.	(CO-1)	6
Q.2 Solve the following:		
A) Show that energy is a property of a system. Differentiate between intensive and extensive properties. State any two intensive and extensive properties.	(CO-2)	6
B) A steam at 18 bar and 400°C ($h_1=3251.6$, $v_1 = 0.16849$ m ³ /kg) steadily enters a nozzle whose inlet area is 0.02 m ² . The mass flow rate of steam through nozzle is 5 kg/s. Steam leaves nozzle at 14 bar with a velocity of 275 m/s. The heat loss from nozzle is estimated to be 14 kJ/s. Determine (a) the inlet velocity of steam (b) specific enthalpy of steam at exit.	(CO-2)	6
Q.3 Solve the following:		
A) Draw schematic diagram of a simple steam power plant. Explain the Carnot heat engine cycle, with the help of T-s diagram with reference to saturation curve, for a simple steam power plant.	(CO-2)	6
B) An inventor claims to have developed an engine that takes in 105 MJ at a temperature of 400 K, rejects 42 MJ at a temperature of 200 K, and delivers 15 kWh of work. Check the validity of the inventors claim. Represent this heat engine with the help of simple schematic diagram.	(CO-2)	6
Q.4 Solve Any ONE of the following:		
A) Show that the adiabatic mixing on two fluids is irreversible process. Two kg of water at 80°C is mixed adiabatically with three kg of water at 30°C at atmospheric pressure. Find increase in entropy due to mixing process. State the assumptions made, if any.	(CO-2)	12
B) Show that heat transfer process through a finite temperature difference is irreversible. Water is heated at a constant pressure of 7 bar. The boiling point of water is 165°C. The initial temperature of water is 0°C. The latent heat of evaporation is 2066.3 kJ/kg. Find the increase of entropy of water, if the final state is dry saturated steam.	(CO-2)	12

Q. 5 Solve the following:

Exhaust gases from a gas turbine are used to heat water in an adiabatic counter-flow heat exchanger. The gases are cooled from 260°C to 120°C, while water enters at 65°C. The flow-rates of gas and water are 0.38 kg/s and 0.5 kg/s respectively. Calculate gain in unavailable energy (exergy loss) due to heat transfer process (Take: $C_{p_g} = 1.09 \text{ kJ/kg K}$, $C_{p_w} = 4.187 \text{ kJ/kg K}$, $T_0 = 25^\circ\text{C}$) (CO-4) **12**

Q. 6 Solve the following:

- A) A closed system of 2 kg of air initially at pressure of 5 bar and 227°C expands polytropically to a pressure of 2 bar following the law of $PV^{1.25} = \text{constant}$. (CO-3) **6**
- (a) Determine work done & heat transfer.
- (b) State assumptions made.
- B) Explain, with the help of simple sketch, the principle of throttling calorimeter. (CO-5) **6**
- Explain with the help of h-s diagram, why a throttling calorimeter cannot measure the quality of very wet steam. How is the quality of such a steam measured then?

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE -
RAIGAD -402 103
Semester Examination – May/June - 2019

Branch: Mechanical Engineering

Sem.: - III

Subject with Subject Code:- Thermodynamics

Marks: 60

Date:- 01/06/2019

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

(Marks)

Q.1. a) Define thermodynamic system. Give classification of thermodynamic system. (06)

b) Explain forms of work in detail. (06)

Q.2. a) State and derive steady flow energy equation and apply to nozzle. (06)

b) In a steady flow system 50 KJ of work is done per kg of fluid, the values of specific volume, pressure and velocity at the inlet and exit sections are $0.4 \text{ m}^3/\text{kg}$, 600 KPa and 15 m/s and $0.6 \text{ m}^3/\text{kg}$, 100 kPa and 250 m/s respectively. The inlet is 30 m above exit. The heat loss from the system is 8 KJ/kg. Calculate the change in internal energy per kg of fluid. (06)

Q.3. a) Prove that efficiency of Carnot cycle is given by Carnot Cycle efficiency $(T_1 - T_2 / T_1)$ (06)

b) A reversible heat engine operates with two environments. In the first it draws 12000 kW from a source at 400°C and in the second it draws 25000 kW from a source at 100°C . In both the operations the engine

rejects heat to a thermal sink at 20 °C. Determine the operation in which the engine delivers more power. (06)

Q.4. a) Show that entropy is property of system. (06)

b) How much entropy of 5kJ of ice will change as it melts into water at the temperature? Assume latent heat of fusion of ice as 335 kJ/kg at 0°C. (06)

Q.5. a) Explain i) Availability ii) Unavailability iii) Dead state (06)

b) How much of the 1200 kJ of thermal energy at 700 K can be converted to useful work if the environment is at 25°C. (06)

Q.6. a) Draw following charts for steam i) T-S and ii) h-s (03)

b) Explain steam table and their uses (03)

c) A closed system containing 1 kg of air at 10 bar and 227°C expands adiabatically to a pressure of 1 bar. Assuming air as an ideal gas ($R=287$ J/kgK, $\gamma=1.4$)

i) calculate volume of air before and after expansion.

ii) Determine work done and heat transfer. (06)

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE - RAIGAD - 402 103
Winter Semester Examination - Dec. - 2019

Branch: B. Tech in Mechanical Engineering
Subject: Thermodynamics (BTME305)
Date: 19/12/2019

Sem.: - III
Marks: 60
Time: - 3 Hr.

Instructions to the Students

1. Each question carries 20 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.
5. Use of steam tables is allowed.

Q.1. Attempt the following:

- (A) Separate the list P (pressure), V (volume), v (specific volume), ρ (density), m (mass), and t (temperature) into intensive properties, and extensive properties. **(03)**
- (B) What is the zeroth law of thermodynamics? Consider two closed systems A and B. System A contains 3000 kJ of thermal energy at 20°C , whereas system B contains 200 kJ of thermal energy at 50°C . Now the systems are brought into contact with each other. Whether the direction of heat transfer between the two systems will be from system A to system B or from system B to system A? Why? **(03)**
- (C) Determine the total work done by a gas system following an expansion process A-B-C as shown in Figure 1 (C). **(02)**

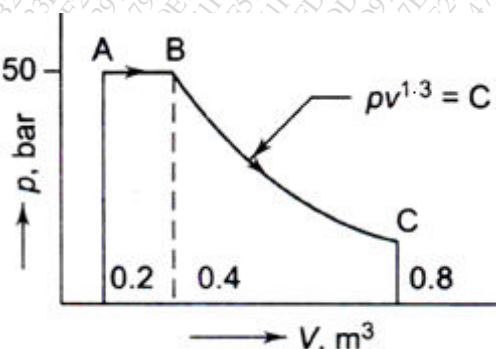


Fig. 1(c). Question 1 (c)

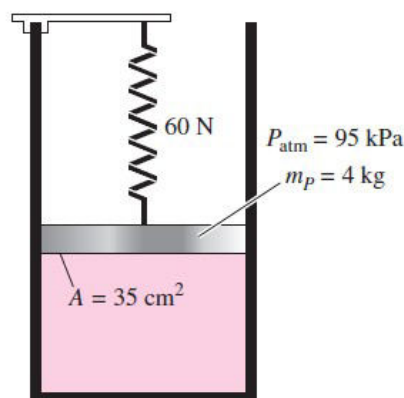


Fig. 1(D), Question 1(D)

(D) A gas is contained in a vertical, frictionless piston-cylinder device (Fig. 1(D)). The piston has a mass of 4 kg and a cross-sectional area of 35 cm². A compressed spring above the piston exerts a force of 60 N on the piston. If the atmospheric pressure is 95 kPa, determine the pressure inside the cylinder in kPa. **(02)**

(E) Explain the following terms: **(02)**

- | | |
|----------------------|---------------------------|
| (i) change of state, | (iii) process |
| (ii) path, | (iv) Thermodynamic cycle. |

Q.2. Attempt the following:

(A) A piston and cylinder machine contains a fluid system which passes through a complete cycle of four processes. During a cycle, the sum of all heat transfers is -170 kJ. The system completes 100 cycles per min. Complete the following table showing the method for each item, and compute the net rate of work output in kW. **(03)**

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)
a-b	0	2170	---
b-c	21000	0	---
c-d	-2100	---	-36600
d-a	---	---	---

(B) A nozzle is a device for increasing the velocity of a steadily flowing stream. At the inlet to a certain nozzle, the enthalpy of the fluid passing is 3000 kJ/kg and the velocity is 60 m/s. At the discharge end, the enthalpy is 2762 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it.

(a) Find the velocity at exits from the nozzle in m/s. **(02)**

(b) If the inlet area is 0.1 m² and the specific volume at inlet is 0.187 m³/kg, find the mass flow rate in kg/s. **(01)**

(c) If the specific volume at the nozzle exit is 0.498 m³/kg, find the exit area of the nozzle. **(02)**

(C) What is the difference between 'work transfer' and 'heat transfer'? What is 'point function' and 'path function'? **(04)**

Q.3. Attempt the following

- (A) What is cyclic heat engine? Define thermal efficiency of a heat engine. Can it be 100? Why? **(03)**
- (B) What is the difference between a heat pump and a refrigerator? Show that COP of a heat pump is greater than COP of a refrigerator by unity. **(04)**
- (C) What is PMM-1 and PMM-2? **(01)**
- (D) Using an engine of 30% thermal efficiency to drive a refrigerator having a COP of 5, what is the heat input into the engine for each MJ removed from the cold body by the refrigerator? If this system is used as a heat pump, how many MJ of heat would be available for heating for each MJ of heat input to the engine? **(04)**

Q.4. Attempt the following:

- (A) What are the four processes that make up the Carnot cycle? Show the Carnot cycle on P-v and T-s diagrams. What is reversed Carnot cycle? What do you understand by Irreversibility and 'inequality of Clausius'? **(02)**
- (B) During the isothermal heat addition process of a Carnot cycle, 900 kJ of heat is added to the working fluid from a source at 400°C. Determine (a) the entropy change of the working fluid, (b) the entropy change of the source, and (c) the total entropy change for the process. **(05)**

Q.5. Attempt the following:

- (A) Explain the following terms giving suitable example. **(06)**
- Available energy and unavailable energy
 - High grade energy and low grade energy
 - Dead state
- (B) Calculate the decrease in available energy when 25 kg of water, at 95°C mixes with 35 kg of water at 35°C, the pressure being taken as constant and the temperature of the surroundings being 15°C. (Cp of water = 4.2 kJ/kg K). **(06)**

Q.6. Attempt the following:

- (A) Steam enters an engine at a pressure 10 bar absolute and 400°C. It is exhausted at 0.2 bar. The steam at exhaust is 0.9 dry. Find : **(02)**
- change in enthalpy ;
 - Change in entropy.
- (B) A throttling calorimeter is used to measure the dryness fraction of the steam in the steam main which has steam flowing at a pressure of 8 bar. The steam after passing through the calorimeter is at 1 bar pressure and

115°C. Calculate the dryness fraction of the steam in the main.

(03)

(C) 0.1 m³ of an ideal gas at 300 K and 1 bar is compressed adiabatically to 8 bar. It is then cooled at constant volume and further expanded isothermally so as to reach the condition from where it started. Calculate:

(i) Pressure at the end of constant volume cooling.

(01)

(ii) Change in internal energy during constant volume process in kJ.

(02)

(iii) Net work done and heat transferred during the cycle. **(04)**

Assume $C_p = 14.3 \text{ kJ/kg K}$ and $C_v = 10.2 \text{ kJ/kg K}$.

*****Paper End*****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch :Mechanical Engg. Semester :III Subject Code & Name: BTMC303 Thermodynamics Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		12
A)	Explain closed and open system with examples.	CO1	6
B)	Derive an equation of work done for constant temperature process in a closed system.	CO1	6
C)	Explain Quasi Static process with a neat sketch.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	State and explain first law of thermodynamics for a closed system undergoing a cycle and process.	CO2	6
B)	Prove that the Energy – A property of system (Point function).	CO2	6
C)	In an air compressor air flows steadily at the rate of 0.5 kg/s through an air compressor. It enters the compressor at 6 m/s with a pressure of 1 bar and a specific volume of 0.85 m ³ /kg and leaves at 5 m/s with a pressure of 7 bar and a specific volume of 0.16 m ³ /kg. The internal energy of the air leaving is 90 kJ/kg greater than that of the air entering. Cooling water in a jacket surrounding the cylinder absorbs heat from the air at the rate of 60 kJ/s. Calculate : (i) The power required to drive the compressor ; (ii) The inlet and output pipe cross-sectional areas.	CO2	6
Q.3	Solve Any Two of the following.		12
A)	State and explain the Kelvin- Plank and Clausius statements of second law of thermodynamic.	CO3	6
B)	Define the Entropy. Explain the Clausius inequality equations.	CO3	6
C)	A Carnot cycle operates between source and sink temperatures of 250°C and – 15°C. If the system receives 90 kJ from the source, find: (i) Efficiency of the system (ii) The net work transfer (iii) Heat rejected to sink.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	State and explain Boyle's law, Charl's law and Gay-Lussac law.	CO4	6
B)	Derive the relation: $C_p - C_v = R$ Where C_p =Specific heat at constant pressure C_v = Specific heat at constant volume R =Characteristic Gas Constant	CO4	6

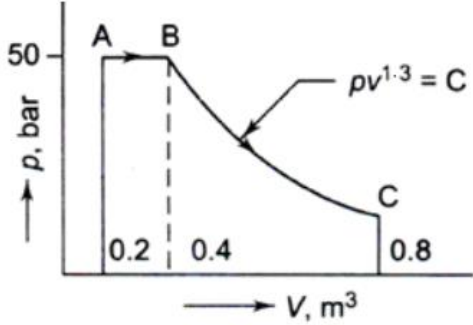
C)	One kg of ideal gas is heated from 18.3°C to 93.4°C. Assuming $R=0.287$ kJ/kg.k and $\gamma =1.18$ for the gas, find out: (i) Specific heats (ii) change in internal energy (iii) Change in enthalpy	CO4	6
Q.5	Solve Any Two of the following.		12
A)	Draw and explain a p-T (pressure-temperature) diagram for a pure substance.	CO5	6
B)	Define (i) Triple point (ii) Dryness fraction of steam (iii) Sensible heat and Latent heat.	CO5	6
C)	Write a short note on Mollier diagram.	CO5	6
	*** End ***		

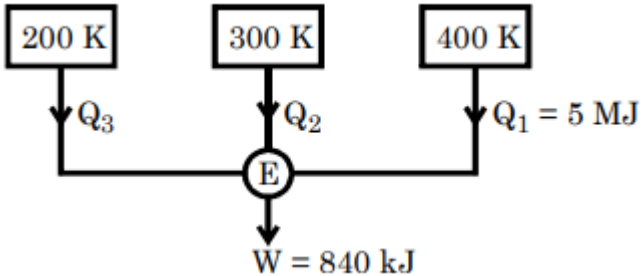
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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary End Semester Examination – Winter 2022-23 Course: B. Tech. Branch: Mechanical Engg. Semester : III Subject Code & Name: Thermodynamics (BTMEC305) Max Marks: 60 Date: __/01/2023 Duration: 3 Hr.															
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly. 5. Use of steam table is permitted.															
Q. 1	Attempt the following	(Level/CO)	Marks												
A)	Describe various types of thermodynamic equilibrium.	Understand	4												
B)	In a steam power plant, steam flows steadily from the steam generator to the turbine. In order to increase the velocity of steam it passes through a horizontal nozzle having an inlet area of 0.1 m². Steam at 1000 kPa and 400°C enters the nozzle with a velocity of 70 m/s. The steam leaves the nozzle at 300 kPa and 300°C and there is a heat loss of 30 kW while steam passing through it. Properties of steam can be referred from the table: <table border="1"> <thead> <tr> <th>Pressure (kPa)</th><th>Temperature (°C)</th><th>Enthalpy (kJ/kg)</th><th>Specific Volume (m³/kg)</th></tr> </thead> <tbody> <tr> <td>1000</td><td>400</td><td>3264</td><td>0.30</td></tr> <tr> <td>300</td><td>300</td><td>3069.3</td><td>0.88</td></tr> </tbody> </table> Determine the mass flow rate of steam, the exit velocity and the exit area of the nozzle.	Pressure (kPa)	Temperature (°C)	Enthalpy (kJ/kg)	Specific Volume (m ³ /kg)	1000	400	3264	0.30	300	300	3069.3	0.88	Evaluate	8
Pressure (kPa)	Temperature (°C)	Enthalpy (kJ/kg)	Specific Volume (m ³ /kg)												
1000	400	3264	0.30												
300	300	3069.3	0.88												
Q.2	Attempt any two questions A/B/C														
A)	Describe the three laws of thermodynamics: 1) Zeroth Law, 2) First law and 3) Statements of second law of thermodynamics.	Understand	6												
B)	Describe the following terms: 1) Types of system, 2) Control Volume, 3) Homogeneous and heterogeneous System, 4) Extensive and Intensive properties, along with practical examples.	Understand	6												
C)	A rigid tank contains 1 m ³ of air at 200 kPa and at 50°C. How much heat must be transferred to the air to raise the air pressure to 400 kPa? Air can be considered as an ideal gas and the value of constant volume specific heat (C _v) for air at 210°C is 0.742 kJ/kg °C.	Analyze and Evaluate	6												
Q. 3	Attempt any two of the following														
A)	A cyclic heat engine operates between a source temperature of 800°C and a sink temperature of 30°C. What is the least rate of heat rejection for 1 kW as net work output of the engine?	Evaluate	6												
B)	Describe the Inequality of Clausius and the three criteria/conditions of the reversibility of a cycle.	Remember /Understand	6												
C)	Describe the applications of entropy principle. Also describe the phenomenon of entropy transfer with the heat flow.	Evaluate	6												
Q.4	Attempt any two of the following														
A)	Describe the terminology: i) High and Low grade energy, ii) Available and Unavailable energy, iii) Dead state	Understand	6												

B)	What is Quality or Dryness fraction of steam. Describe any one calorimeter in detail which can measure the quality of steam, with the help of neat schematic and mathematical expressions.	Understand	6
C)	Steam initially at 0.3 MPa, 250°C is cooled at constant volume. (i) at what temperature will the steam become saturated vapour ? (ii) What is the quality at 80°C ? What is the heat transferred per kg of steam in cooling from 250°C to 80°C.	Remember	6
Q. 5	Attempt the following		
A)	A fluid at 200 kPa and 300°C has a volume of 0.8 m ³ . In a frictionless process at constant volume the pressure changes to 100 kPa. Find the final temperature and the heat transferred if the fluid is air. Also find the properties if the fluid is steam.	Analyze and Evaluate	6
B)	State and describe the various process steam can undergo in a cycle with the help of P-h (Pressure –Enthalpy) and or T-S (Temperature-Entropy) Charts.	Understand	6
*** End ***			

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2023 Course: B. Tech. Branch : Mechanical Engineering Semester :III Subject Code & Name: Thermodynamics (BTMC303) Max Marks: 60 Date: 14-08-2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Use of steam table is allowed 5. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Mar ks
Q. 1	Solve Any Two of the following.		12
A)	Define the thermodynamic system. Classify it with brief explanation and example of each.	L2 - Understand CO1	6
B)	Define thermodynamic equilibrium and classify it with brief explanation of each	L2 - Understand CO1	6
C)	Determine the total work done by a gas system following an expansion process as shown in Figure. 	L3 -Apply CO1	6
Q.2	Solve Any Two of the following.		12
A)	A mass of 8 kg gas expands within a flexible container so that the p–V relationship is of the form $pV^{1.2} = \text{constant}$. The initial pressure is 1000 kPa and the initial volume is 1 m³ . The final pressure is 5 kPa . If specific internal energy of the gas decreases by 40 kJ/kg , find the heat transfer in magnitude and direction.	L3 -Apply CO2	6
B)	A reciprocating air compressor takes in 2 m³/min at 0.11 MPa, 20°C which it delivers at 1.5 MPa, 111°C to an aftercooler where the air is cooled at constant pressure to 25°C . The power absorbed by the compressor is 4.15 kW . Determine the heat transfer in (a) The compressor (b) The cooler. State your assumptions.	L3 -Apply CO2	6

C)	Explain different forms of energy and write down the equation of each.	L2 - Understand CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Two reversible heat engines A and B are arranged in series, A rejecting heat directly to B. Engine A receives 200 kJ at a temperature of 421°C from a hot source, while engine B is in communication with a cold sink at a temperature of 4.4°C . If the work output of A is twice that of B, find (a) The intermediate temperature between A and B (b) The efficiency of each engine (c) The heat rejected to the cold sink	L3 -Apply CO3	6
B)	A reversible engine, as shown in Figure during a cycle of operations draws 5 MJ from the 400 K reservoir and does 840 kJ of work. Find the amount and direction of heat interaction with other reservoirs. 	L3 -Apply CO3	6
C)	State and explain the following statements of second law of thermodynamics 1. Clausius Statement of the Second Law of thermodynamics. 2. Kelvin–Planck Statement of the Second Law of thermodynamics.	L2 - Understand CO3	6
Q.4	Solve Any Two of the following.		12
A)	Air contained in a cylinder fitted with a piston is compressed reversibly according to the law $pV^{1.25} = \text{const}$. The mass of air in the cylinder is 0.1 kg . The initial pressure is 100 kPa and the initial temperature 20°C . The final volume is 1/ 8 of the initial volume. Determine the work and the heat transfer.	L3 -Apply CO4	6
B)	A closed system allows nitrogen to expand reversibly from a volume of 0.25 m³ to 0.75 m³ along the path $pV^{1.32} = \text{const}$. The original pressure of the gas is 250 kPa and its initial temperature is 100°C. (a) Draw the p-v and T-s diagrams. (b) What are the final temperature and the final pressure of the gas? (c) How much work is done and how much heat is transferred? (d) What is the Entropy change of nitrogen?	L3 -Apply CO4	6
C)	State the following laws related to ideal gases and write the their equations 1. Boyle's Law 2. Charles' Law 3. Gay Lussac's Law 4. Avogadro's Law	L2 - Understand CO4	6

Q. 5	Solve Any Two of the following.		12
A)	Explain the phase change process of the pure substance and plot the same on T-s Plot by assume the initial state of the substance to be solid ice and it is heated at constant pressure.	L2 - Understand CO5	6
B)	Define the following terms related to pure substances (only definitions) 1. Saturated Liquid. 2. Saturated Vapor States 3. Saturated Liquid–Vapor Mixture 4. Superheated Vapor 5. Compressed Liquid 6.Dryness Fraction	L2 - Understand CO5	6
C)	Find the enthalpy and entropy of steam when the pressure is 2 MPa and the specific volume is 0.09 m³/kg .	L3 -Apply CO5	6
	*** End ***		

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Instructions to the Students:

1. All the questions are compulsory.
2. Attempt any two sub questions from each question
3. Use of non-programmable scientific calculators is allowed.
4. Illustrate your answer with neat sketches, diagram, etc., wherever necessary.
5. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

A) Clearly differentiate between the following with suitable examples

Remembering

6

- i) Control Mass and Control Volume
- ii) Microscopic and Macroscopic
- iii) Path function and point function

B) i) A barometer measures 760 mm Hg at street level and 735 mm Hg on top of a building. How tall is the building if we assuming air density of 1.15 kg/m³.

Analyzing

6

ii) The limiting value of the ratio of the pressure of gas at the steam point and at the triple point of water when the gas is kept at constant volume is found to be 1.36605. What is the ideal gas temperature of steam point.

C) i) State the zeroth law of thermodynamics and define thermometric property.

Understanding

6

ii) Obtain the expression for quasi static pdv work for isothermal process.

Q.2 Solve Any Two of the following.

12

A) State first Law of thermodynamics for a cyclic process and show that it is consistent with the following situation.

Remembering

6

Process	Q (kJ)	W (kJ)
1-2	443	255
2-3	- 412	-340
3-4	- 222	0
4-1	255	149

B) Started from first principle show that throttling is an isenthalpic process by apply SFEE.

Understanding

6

C) Air flows steadily at the rate of 0.4 kg/s through an air compressor, entering at 6 m/s with a pressure of 1 bar and a specific volume of 0.85 m³/kg and

Analyzing

6

leaving at 4.5 m/s with a pressure of 6.9 bar and a specific volume of 0.16 m³/kg. The internal energy of the air leaving is 88 kJ/kg greater than that of the air entering. Cooling water jackets surrounding the cylinder absorbs heat from the air at the rate of 59 W.

Q. 3 Solve Any Two of the following.		12
A) Define Heat Pump and refrigerator with suitable diagram and show that $COP_{HP} = COP_{Ref} + 1$	Understanding	6
B) State the entropy principal and explain the entropy balance and with proper illustration and define the concept of entropy generation.	Understanding	6
C) A heat pump is to be used to heat a house in winter and then reversed to cool the house in summer. The interior temperature is to be maintained at 20 °C. Heat transfer through the walls and roof is estimated to be 0.525 kJ/s per degree temperature difference the inside and outside. If the outside temperature in winter is 5 °C, what is the minimum power required to drive the heat pump.	Analyzing	6
Q.4 Solve Any Two of the following.		12
A) Define ideal gas. What is Universal gas constant? Obtain its value.	Remembering	6
B) A mass of 1 kg of air initially at 5 bar and 100 °C undergoes a cycle consisting of constant pressure expansion until the volume is doubled followed by constant volume cooling followed by the reversible adiabatic compression to the initial state. Find the net work done in the cycle.	Analyzing	6
C) Show that for an ideal gas $c_p - c_v = R$.	Analyzing	6
Q. 5 Solve Any Two of the following.		12
A) Explain the working of the throttling calorimeter with the schematic diagram and process representation on h-s diagram.	Remembering	6
B) Steam initially at 1.5 MPa, 300 °C expands reversibly and adiabatically in a steam turbine to 40 °C. Determine the ideal work output of the turbine per kg of steam.	Analyzing	6
C) Define the following; Dryness Fraction, saturated states, critical states	Remembering	6

End of the Manuscript